Non-stationarity and unit-root tests

Econometrics. Lecture 9

Content

1. Non-stationarity
2. Consequences of non-stationarity
3. Testing for non-stationarity
4. ARIMA models estimation and diagnostic

1. Non-stationarity

Recap

- **Covariance stationarity requires**:
  - constant mean;
  - constant variance;
  - covariance that depends only on the lag j between two observations, not on where they sit in time:

$$Cov(Y_t, Y_{t+j}) = Cov(Y_s, Y_{s+j}) = \gamma_j$$

- If any of these conditions fails, the series is **non-stationary.**

What fails when a series is non-stationary

– **No level to return to.** There is no unconditional mean in the model, so nothing corresponds to a long-run equilibrium value, and a long-horizon forecast has nothing to converge on.

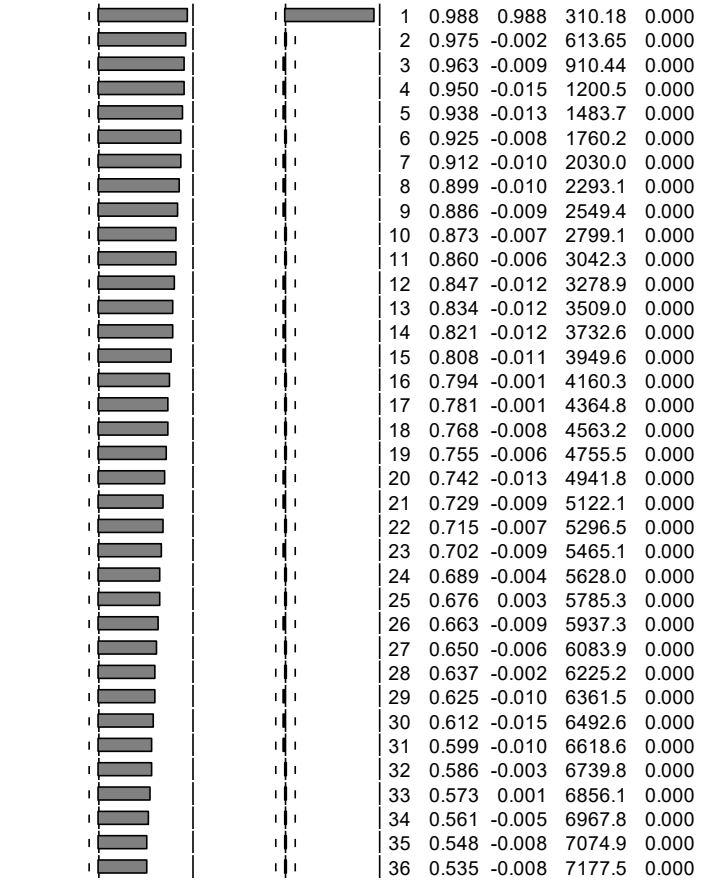
– **A variance that grows with the sample.** The uncertainty around a forecast widens without limit as the horizon extends.

– **Memory that does not fade.** Theoretical autocorrelations do not decay, sample autocorrelations decay only very slowly.

Which variable is non-stationary?

Time series 1

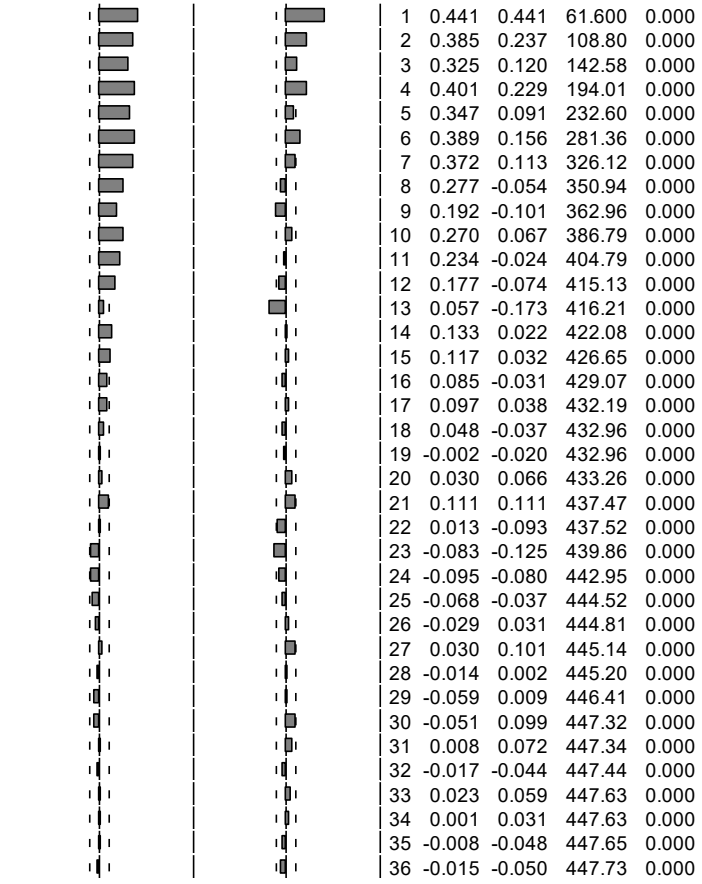
Date: 05/07/26 Time: 13:42
Sample: 2000M01 2026M03
Included observations: 315



Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.988	0.988	310.18	0.000
		2	0.975	-0.002	613.65	0.000
		3	0.963	-0.009	910.44	0.000
		4	0.950	-0.015	1200.5	0.000
		5	0.938	-0.013	1483.7	0.000
		6	0.925	-0.008	1760.2	0.000
		7	0.912	-0.010	2030.0	0.000
		8	0.899	-0.010	2293.1	0.000
		9	0.886	-0.009	2549.4	0.000
		10	0.873	-0.007	2799.1	0.000
		11	0.860	-0.006	3042.3	0.000
		12	0.847	-0.012	3278.9	0.000
		13	0.834	-0.012	3509.0	0.000
		14	0.821	-0.012	3732.6	0.000
		15	0.808	-0.011	3949.6	0.000
		16	0.794	-0.001	4160.3	0.000
		17	0.781	-0.001	4364.8	0.000
		18	0.768	-0.008	4563.2	0.000
		19	0.755	-0.006	4755.5	0.000
		20	0.742	-0.013	4941.8	0.000
		21	0.729	-0.009	5122.1	0.000
		22	0.715	-0.007	5296.5	0.000
		23	0.702	-0.009	5465.1	0.000
		24	0.689	-0.004	5628.0	0.000
		25	0.676	0.003	5785.3	0.000
		26	0.663	-0.009	5937.3	0.000
		27	0.650	-0.006	6083.9	0.000
		28	0.637	-0.002	6225.2	0.000
		29	0.625	-0.010	6361.5	0.000
		30	0.612	-0.015	6492.6	0.000
		31	0.599	-0.010	6618.6	0.000
		32	0.586	-0.003	6739.8	0.000
		33	0.573	0.001	6856.1	0.000
		34	0.561	-0.005	6967.8	0.000
		35	0.548	-0.008	7074.9	0.000
		36	0.535	-0.008	7177.5	0.000

Time series 2

Date: 05/07/26 Time: 13:44
Sample (adjusted): 2000M02 2026M03
Included observations: 314 after adjustments



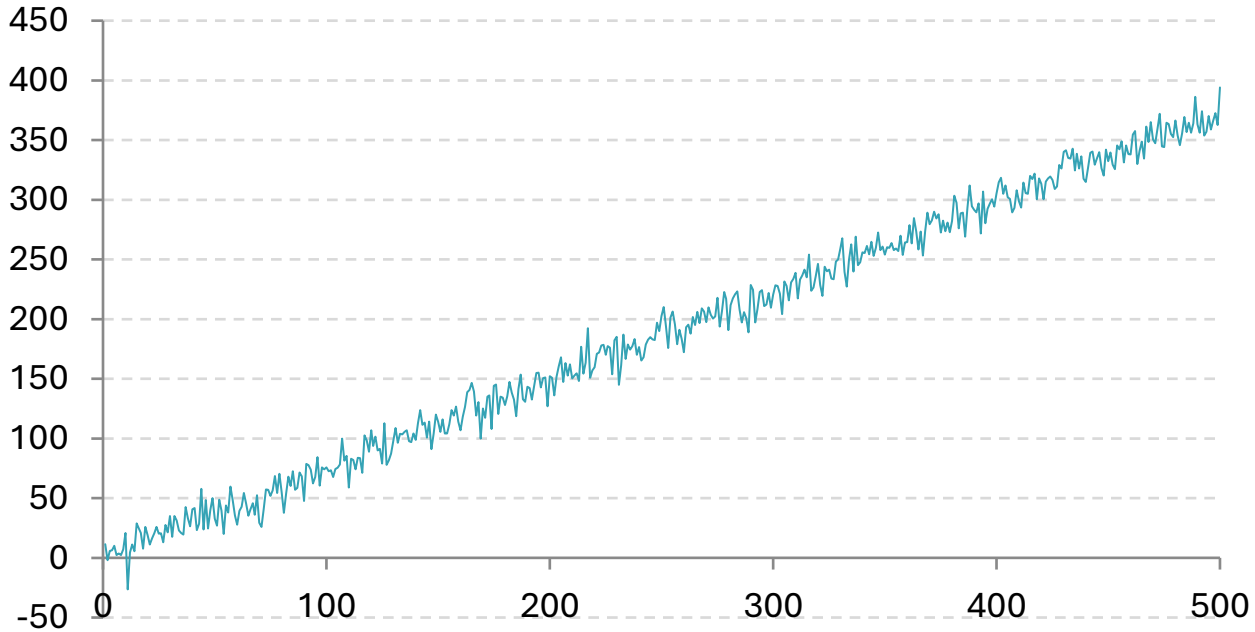
Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.441	0.441	61.600	0.000
		2	0.385	0.237	108.80	0.000
		3	0.325	0.120	142.58	0.000
		4	0.401	0.229	194.01	0.000
		5	0.347	0.091	232.60	0.000
		6	0.389	0.156	281.36	0.000
		7	0.372	0.113	326.12	0.000
		8	0.277	-0.054	350.94	0.000
		9	0.192	-0.101	362.96	0.000
		10	0.270	0.067	386.79	0.000
		11	0.234	-0.024	404.79	0.000
		12	0.177	-0.074	415.13	0.000
		13	0.057	-0.173	416.21	0.000
		14	0.133	0.022	422.08	0.000
		15	0.117	0.032	426.65	0.000
		16	0.085	-0.031	429.07	0.000
		17	0.097	0.038	432.19	0.000
		18	0.048	-0.037	432.96	0.000
		19	-0.002	-0.020	432.96	0.000
		20	0.030	0.066	433.26	0.000
		21	0.111	0.111	437.47	0.000
		22	0.013	-0.093	437.52	0.000
		23	-0.083	-0.125	439.86	0.000
		24	-0.095	-0.080	442.95	0.000
		25	-0.068	-0.037	444.52	0.000
		26	-0.029	0.031	444.81	0.000
		27	0.030	0.101	445.14	0.000
		28	-0.014	0.002	445.20	0.000
		29	-0.059	0.009	446.41	0.000
		30	-0.051	0.099	447.32	0.000
		31	0.008	0.072	447.34	0.000
		32	-0.017	-0.044	447.44	0.000
		33	0.023	0.059	447.63	0.000
		34	0.001	0.031	447.63	0.000
		35	-0.008	-0.048	447.65	0.000
		36	-0.015	-0.050	447.73	0.000

Trend stationary processes (TS)

- **Deterministic trend:** nonrandom *function of time*:

$$y_t = \mu + \beta t + u_t, \quad \text{where } t \text{ is time and } u_t \text{ is a white noise}$$

$\beta = 0.75$



Difference stationary processes (DS)

- **Stochastic trend:** random trend, varies over time
  - Random walk (RW) $y_t = y_{t-1} + u_t$

    u_t is a white noise
  - Random walk with drift $y_t = \mu + y_{t-1} + u_t$
    - μ is the "drift"; if $\mu > 0$ the series grows continuously

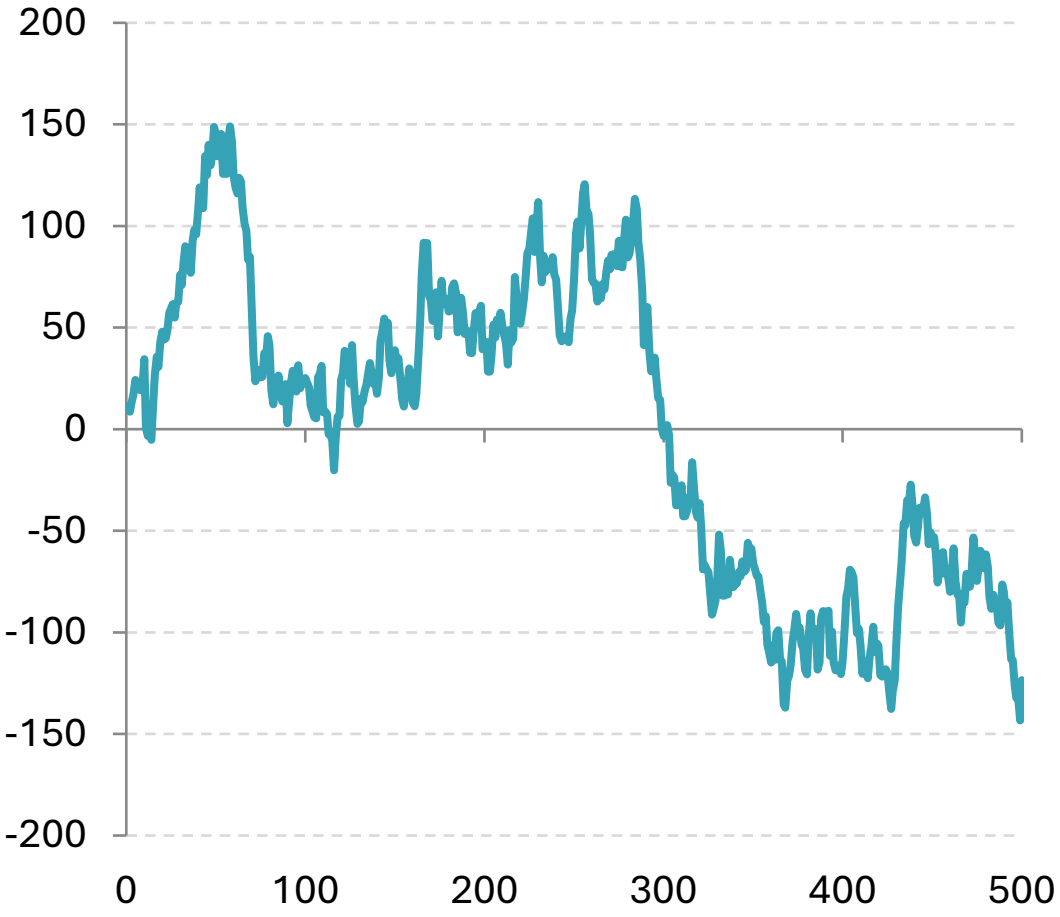
Terminology

• Order of integration: number of times a series y must be differenced to become stationary

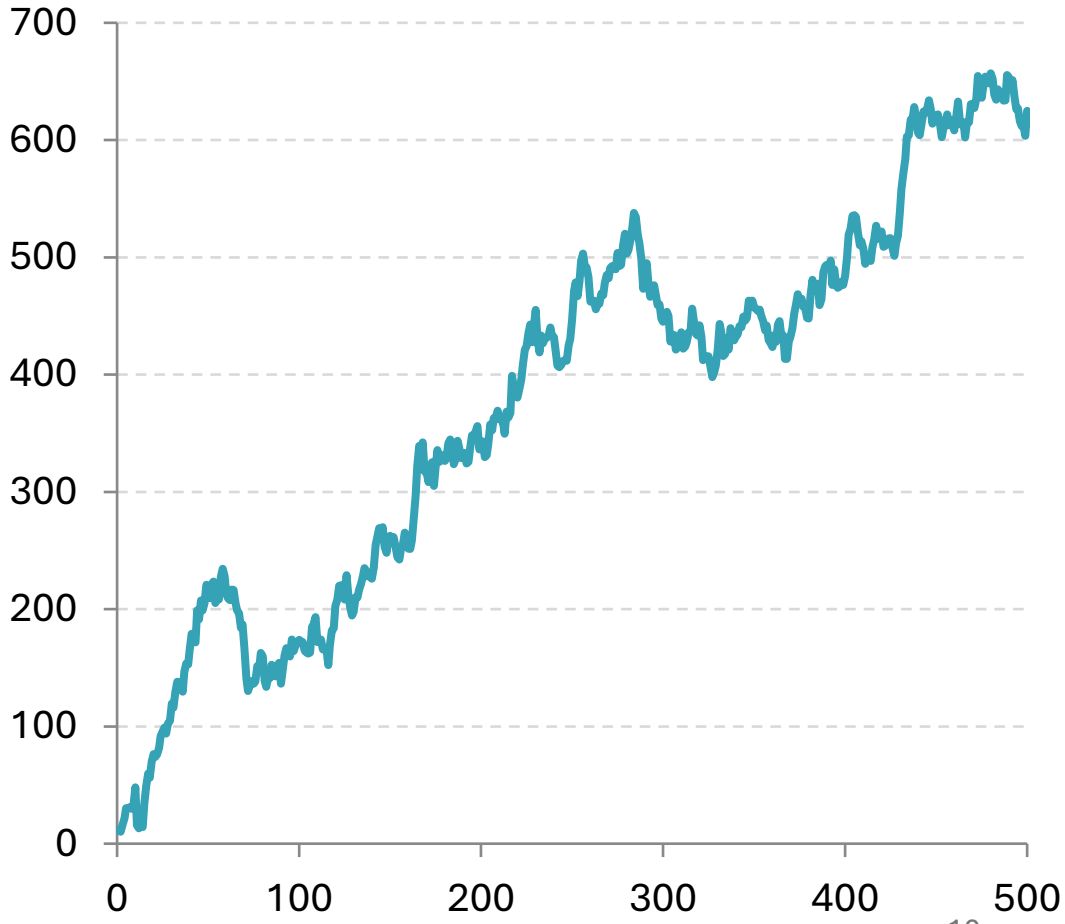
Notation	Meaning
I(0)	Stationary as it stands – no differencing needed. Stationary in levels.
I(1)	The first difference Δy is stationary; y itself is not.
I(2)	The difference of the difference is stationary; neither y nor Δy is

Random walks

Random walk

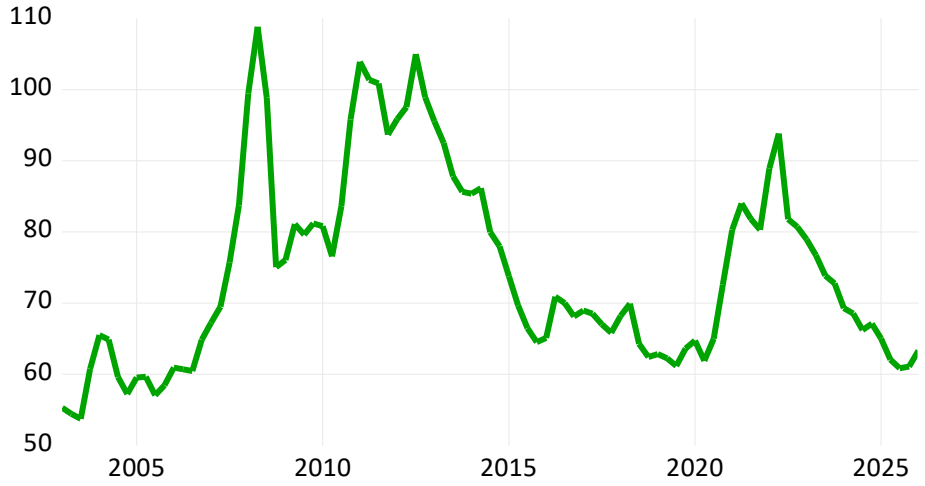


Random walk with drift ($\mu = 1.5$)

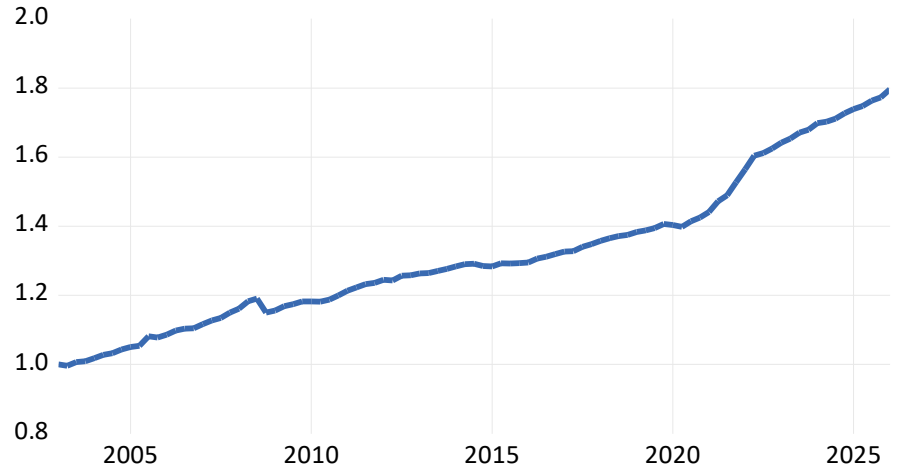


What process?

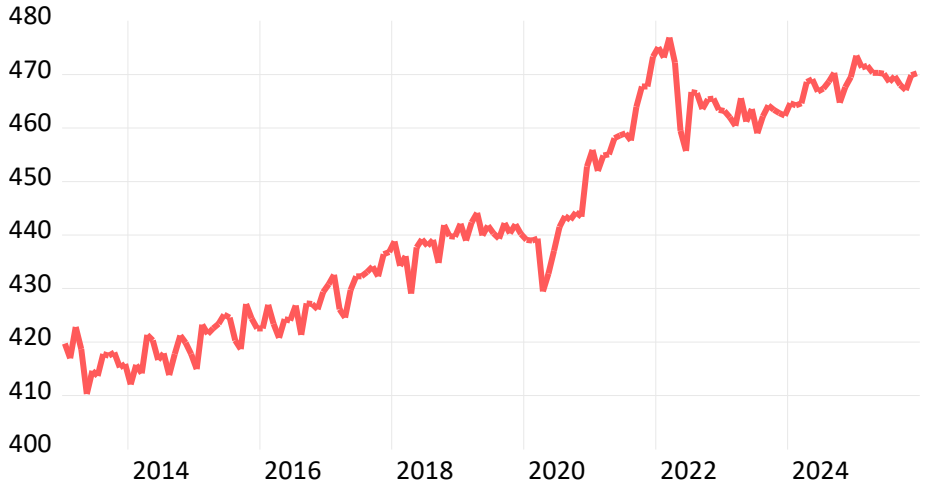
Time series 1



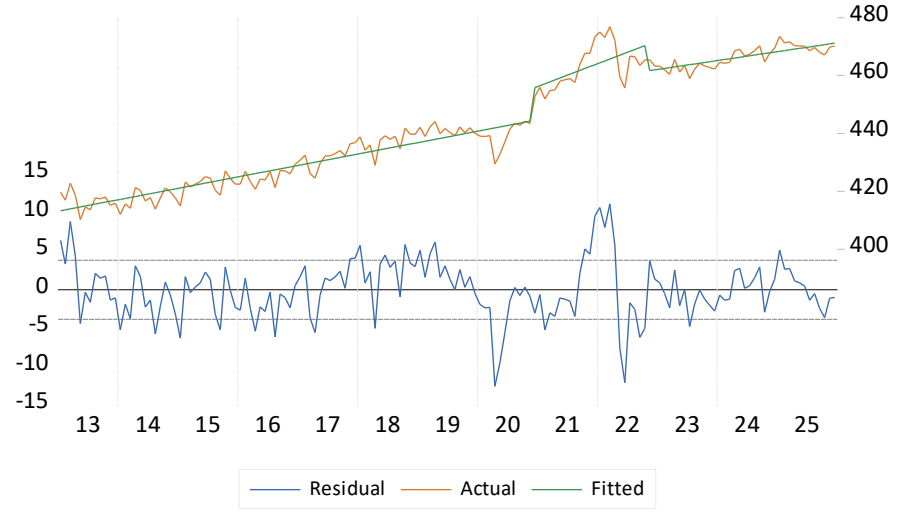
Time series 2



Time series 3



L_MANUF = @BEFORE("2020M12")*(362 + 0.33*@TREND) + @DURING("2020M12 2022M10")*(290 + 0.66*@TREND) + @AFTER("2022M11")*(392 + 0.26*@TREND)



2. Consequences of non-stationarity

Consequences of non-stationarity

- Shocks do not “die out”.

- Test statistics no longer have their usual distributions – they are not normal, so conventional inference fails.

- Autoregressive coefficients are biased, so an AR fitted to a non-stationary series is estimated poorly, which damages forecasts.

- The usual confidence intervals for coefficients are not valid.

Shocks do not “die out” (1)

- Start from a general AR(1):

$$y_t = by_{t-1} + u_t,$$

- and substitute repeatedly for the lagged term. We get an MA(q):

$$y_t = b^t y_0 + u_t + bu_{t-1} + b^2 u_{t-2} + \cdots + b^{t-2} u_2 + b^{t-1} u_1$$

The impact of shocks will depend on the value of b.

Shocks do not “die out” (2)

$$y_t = b^t y_0 + u_t + b u_{t-1} + b^2 u_{t-2} + \cdots + b^{t-2} u_2 + b^{t-1} u_1$$

Three cases:

1. $|b| < 1, b^t \to 0$ as $t \to \infty$, the effect of a shock **diminishes** with time.

2. $b = 1, b^t = 1$ for all t, so the effect of a shock **persists**, variance grows indefinitely with time.

$$y_t = y_0 + \sum_{i=0}^{t-1} u_{t-i}$$

3. $b > 1$, a shock becomes **more influential** over time.

Statistical consequences of non-stationarity

- Test statistics no longer have their usual distributions – they are not normal, so conventional inference fails.
- Autoregressive coefficients are biased, so an AR fitted to a non-stationary series is estimated poorly, which damages forecasts.
- The usual confidence intervals for coefficients are not valid.

Regression between unrelated non-stationary series: Spurious regression

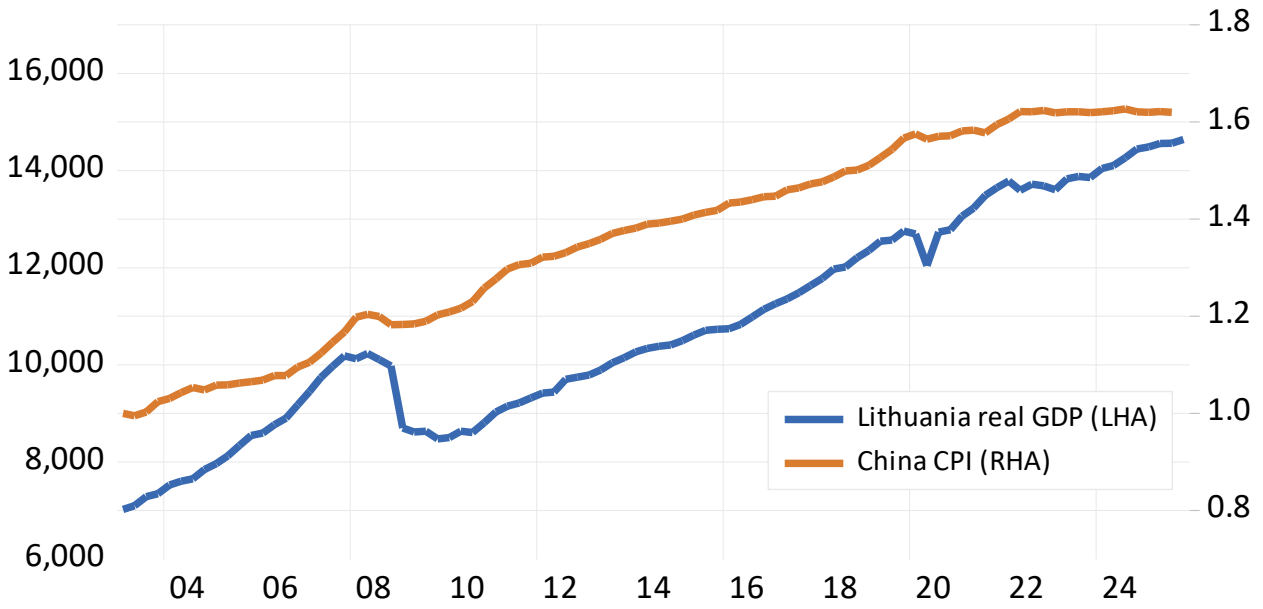
• Two unrelated nonstationary series *y* and *z* might appear to be related through a standard OLS regression:

– an R^2 that drifts upward with the sample rather than settling at zero.

– *t*-statistics that look convincingly significant.

– The true test: are the **regression residuals** stationary? (i.e., long-run equilibrium relationship between *y* and *z*?)

Spurious regression. Lithuania GDP and China CPI



High R^2

Significant *t*-statistic

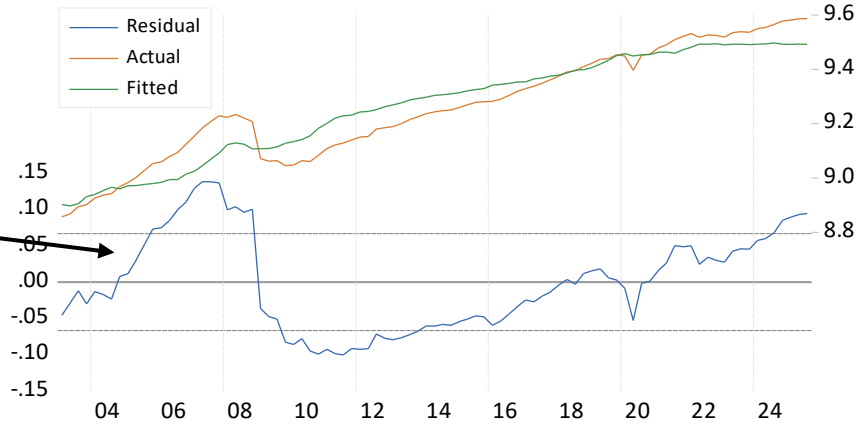
Dependent Variable: LOG(GDP_LT_SA)
Method: Least Squares
Date: 01/18/26 Time: 15:22
Sample (adjusted): 2003Q1 2025Q3
Included observations: 91 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
LOG(CPI_CN)	1.220800	0.045224	26.99479	0.0000
C	8.902736	0.015031	592.2774	0.0000

R-squared	0.891161	Mean dependent var	9.261694
Adjusted R-squared	0.889938	S.D. dependent var	0.201533
S.E. of regression	0.066860	Akaike info criterion	-2.550699
Sum squared resid	0.397853	Schwarz criterion	-2.495515
Log likelihood	118.0568	Hannan-Quinn criter.	-2.528436
F-statistic	728.7189	Durbin-Watson stat	0.088769
Prob(F-statistic)	0.000000		

Residuals are non-stationary.

The regression is spurious.



3. Testing for non-stationarity

Dickey-Fuller (DF) test

- AR(1) model: $y_t = by_{t-1} + u_t$
- Special case: RW, when $b = 1$
- Stationarity requires $|b| < 1$
- AR(1) can be tested for whether *y* is a driftless random walk: $\mathrm{H}_0\colon\ b = 1$

  or, equivalently: $\Delta y_t = \psi y_{t-1} + u_t, \quad \psi = b - 1$

  $\mathrm{H}_0\colon\ \psi = 0 \qquad \mathrm{H}_1\colon\ \psi < 0$

- This is the "Dickey-Fuller" (DF) test:

  – regress Δy_t on its lag, test for significance of the coefficient.

Dickey-Fuller (DF) test. Interpretation

Simple DF test can be extended:

- Intercept: $\Delta y_t = \mu + \psi y_{t-1} + u_t$
- Intercept and time trend: $\Delta y_t = \mu + \psi y_{t-1} + \alpha t + u_t$
- In all three cases, H_0: $\psi = 0$; y has a unit root. And H_1: $\psi < 0$.

Rejecting the null is therefore evidence that the series is stationary.

Critical values for the *t*-statistics of ψ will vary depending on whether intercept, trend are included.

Augmented Dickey-Fuller (ADF) test

DF test assumes that u_t is white noise.

- However, if u_t is autocorrelated, a different version of the test is needed, allowing for higher lags.
- Augmented Dickey-Fuller (ADF) test :

$$\Delta y_t = \{\mu + \alpha t\} + \gamma y_{t-1} + \sum_{i=1}^{p} \beta_i \Delta y_{t-i} + u_t$$

$$\mathrm{H}_0: \gamma = 0 \qquad \mathrm{H}_1: \gamma < 0$$

ADF test

- As with DF, ADF tests whether coefficient on y_{t-1} (γ) = 0

- Two choices the ADF test forces on you

  – **Deterministic terms:** none, intercept, or intercept and trend. Choose on the basis of what the series would look like if it were stationary

  – **Lag length *p***: enough lags to leave the ADF regression's residuals free of autocorrelation, and no more:

    - minimise AIC or SBC over a range of p;
    - start out with large p and drop the longest lag while it remains insignificant.

Situations where DF and ADF tests have trouble

- Stationary processes with near-unit roots

  – $b = 0.95$ and $b = 1$ imply similar behaviour over a hundred observations, and the test has difficulty separating them.

- A break in the deterministic part. A single unmodelled shift in level or slope pushes the test toward non-rejection, and can do so even when the series is trend-stationary throughout.

- Errors with a large negative MA component.

Phillips-Perron test

• Formulation: $\Delta y_t = \mu^* + \delta^* t + \psi y_{t-1} + u_t$, where u_t is I(0) and may be *heteroskedastic* and *autocorrelated*, that is, following an ARMA (*p,q*).

• H_0: $\psi = 0$

• PP corrects for any serial correlation and heteroskedasticity in the errors u_t by directly modifying the test statistics.

• One advantage of PP: no need to specify lag length.

Kwiatkowski–Phillips–Schmidt–Shin (KPSS) test

- Formulation: $y_t = \beta_0 D_t + \mu_t + u_t$, where $\mu_t = \mu_{t-1} + \varepsilon_t$.
- D_t contains deterministic components (constant or constant and time trend).
- μ_t is a random walk.
- H_0: y is stationary or equivalently H_0: $\sigma_\varepsilon^2 = 0$
- H_1: $\sigma_\varepsilon^2 > 0$

Reading tests together

	KPSS does not reject (H_0: stationary)	KPSS rejects
ADF does not reject (H_0: unit root)	Uninformative. Neither hypothesis can be dismissed, usually because the sample is short. Report the ambiguity.	Consistent evidence of a unit root. Difference and proceed.
ADF rejects	Consistent evidence of stationarity. Model in levels.	Conflicting. Suspect a break, a misspecified deterministic term, or a fractionally integrated process.

4. ARIMA models estimation and diagnostic

ARIMA models

ARMA(p,q) models can only be made on time series that are stationary.

If the time series are non-stationary we need to make them stationary. And the traditional approach is to difference the raw data (after taking logs).

$$\Delta y_t = y_t - y_{t-1}$$

If, after first differencing, a series is stationary then the series is called integrated to order one and denoted I(1) – which completes the abbreviation ARIMA.

The general ARIMA model is called an **ARIMA(*p*, *d*, *q*)**, with p being the number of lags of the dependent variable (the AR terms), d being the number of differences required to take in order to make the series stationary, and q being the number of lagged terms of the error term (the MA terms).

The Box–Jenkins approach. Identification stage

Step 1 Visually explore the data. Calculate the ACF and PACF of the raw data, check whether the series is stationary or not. Justify applying tests (ADF, PP, KPSS). If the series is stationary go to step 3, if not go to step 2.

Step 2 Take the logarithm (if possible) and the first differences of the raw data and calculate the ACF and PACF for the first differenced logarithmic series. Apply tests (ADF, PP, KPSS).

Step 3 Examine the graphs of the ACF and PACF and determine which models would be good starting points.

The Box–Jenkins approach. Estimation stage

Step 4 Estimate those models.

– The estimated models are compared using the Akaike information criterion (AIC) and the Schwarz Bayesian criterion (SBC). We want a parsimonious model, so we will choose the model with the smallest AIC and SBC values. Of the two criteria, the SBC is preferable.

The Box–Jenkins approach. Diagnostic stage

Step 5 For each of these estimated models:

– check to see if the parameter of the longest lag is significant. If not, then you probably have too many parameters, and should decrease the order of p and/or q.

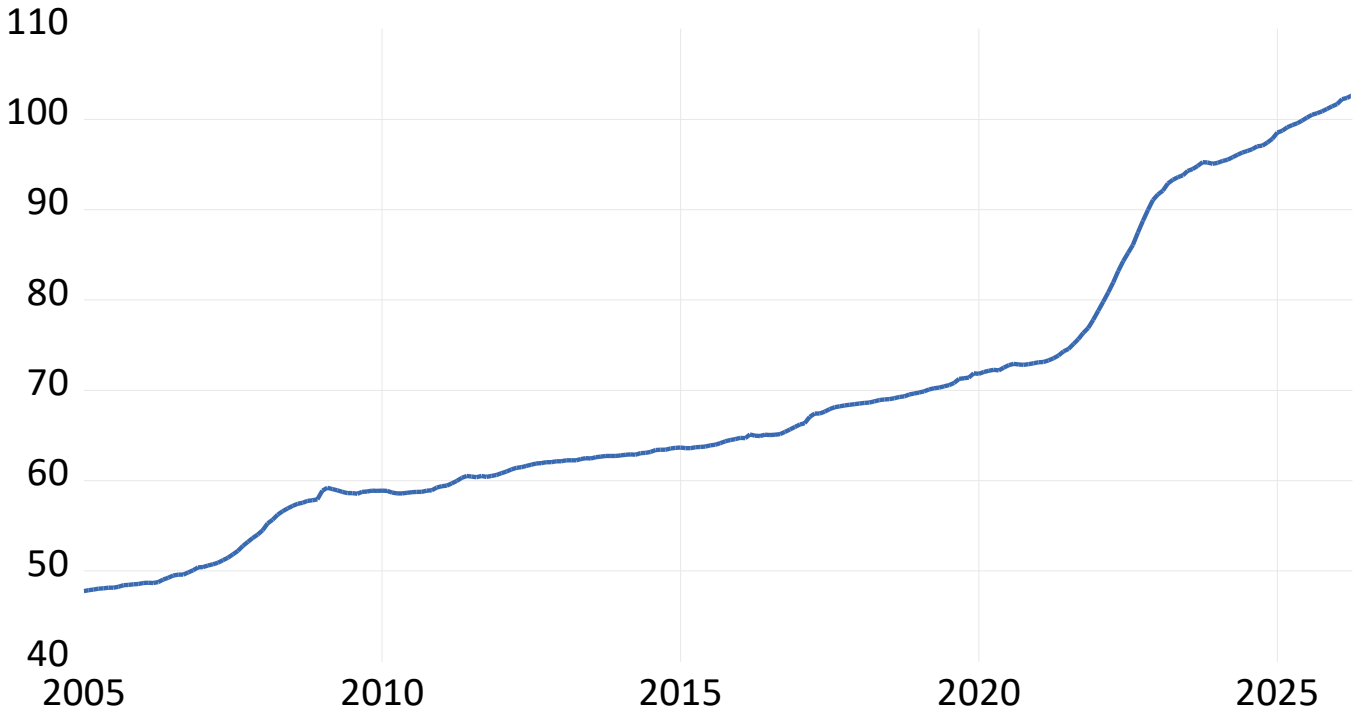
– check the ACF, and PACF of the errors. If the model has at least enough parameters, then all error ACFs and PACFs will be insignificant.

– check the AIC and SBC together with the adjusted R^2 of the estimated models to detect which model is the parsimonious one (i.e. the one that minimizes AIC and SBC and has the highest adjusted R^2).

Step 6 If changes in the original model are needed, go back to step 4.

Lithuania CPI (excl. energy & unproc. food)

Trend in the series – evidence of nonstationarity

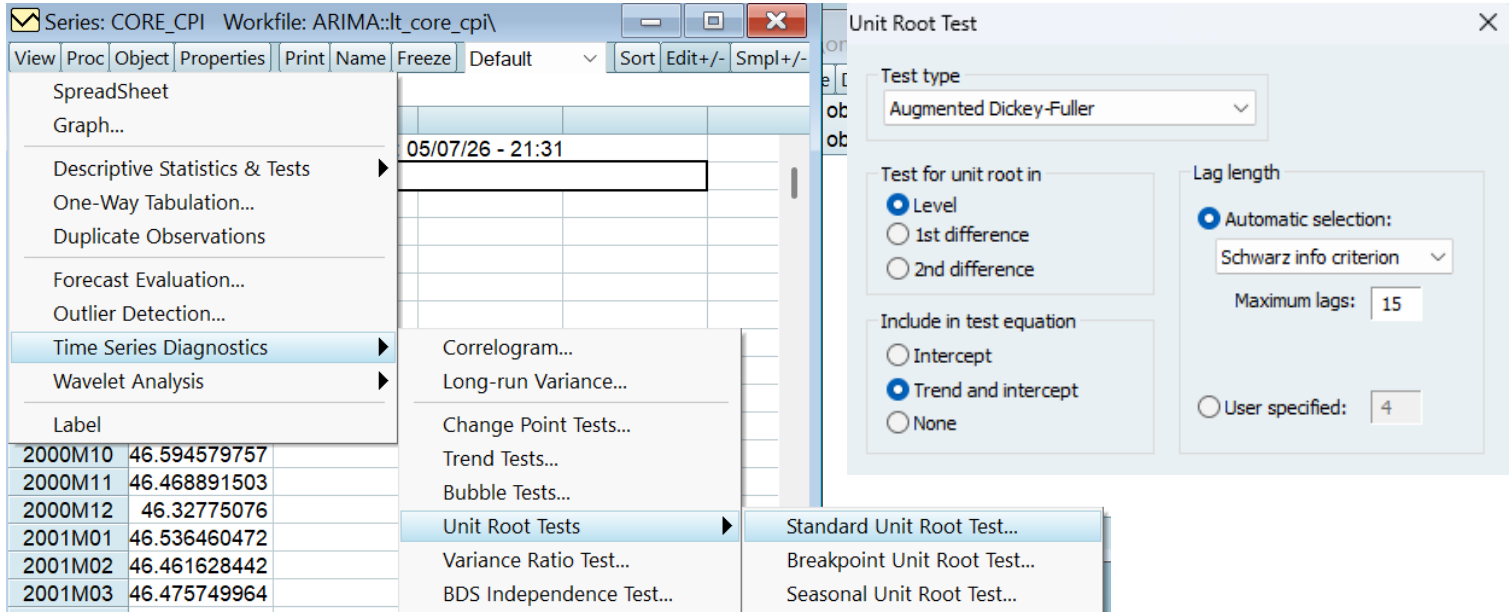


Date: 05/07/26 Time: 21:39
Sample: 2005M01 2026M04
Included observations: 256

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.986	0.986	251.68	0.000
		2	0.971	-0.017	496.95	0.000
		3	0.956	-0.019	735.71	0.000
		4	0.941	-0.010	967.99	0.000
		5	0.926	-0.017	1193.7	0.000
		6	0.911	-0.015	1412.9	0.000
		7	0.895	-0.016	1625.5	0.000
		8	0.879	-0.015	1831.4	0.000
		9	0.863	-0.015	2030.7	0.000
		10	0.847	-0.014	2223.4	0.000
		11	0.831	-0.011	2409.5	0.000
		12	0.814	-0.013	2589.0	0.000
		13	0.798	-0.013	2762.1	0.000
		14	0.781	-0.013	2928.7	0.000
		15	0.765	-0.009	3089.0	0.000
		16	0.748	-0.011	3242.9	0.000
		17	0.732	0.001	3390.8	0.000
		18	0.715	-0.005	3532.8	0.000
		19	0.699	-0.006	3668.9	0.000
		20	0.683	-0.013	3799.4	0.000
		21	0.666	-0.010	3924.2	0.000
		22	0.650	-0.009	4043.5	0.000
		23	0.634	-0.009	4157.3	0.000
		24	0.618	-0.007	4265.9	0.000
		25	0.601	-0.010	4369.3	0.000
		26	0.585	-0.008	4467.7	0.000
		27	0.569	-0.013	4561.1	0.000
		28	0.553	-0.011	4649.7	0.000
		29	0.537	-0.012	4733.5	0.000
		30	0.520	-0.017	4812.5	0.000
		31	0.504	-0.014	4887.0	0.000
		32	0.487	-0.004	4957.1	0.000
		33	0.471	-0.004	5022.8	0.000
		34	0.455	-0.008	5084.5	0.000
		35	0.439	-0.002	5142.2	0.000
		36	0.424	-0.009	5196.1	0.000

ACF does not die down – clearly suggest nonstationarity

Lithuania CPI.
ADF test



Fail to reject the null – series is nonstationary

Null Hypothesis: CORE_CPI has a unit root
Exogenous: Constant, Linear Trend
Lag Length: 3 (Automatic - based on SIC, maxlag=15)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-1.367098	0.8683
Test critical values:	1% level	-3.994310	
	5% level	-3.427476	
	10% level	-3.137059	

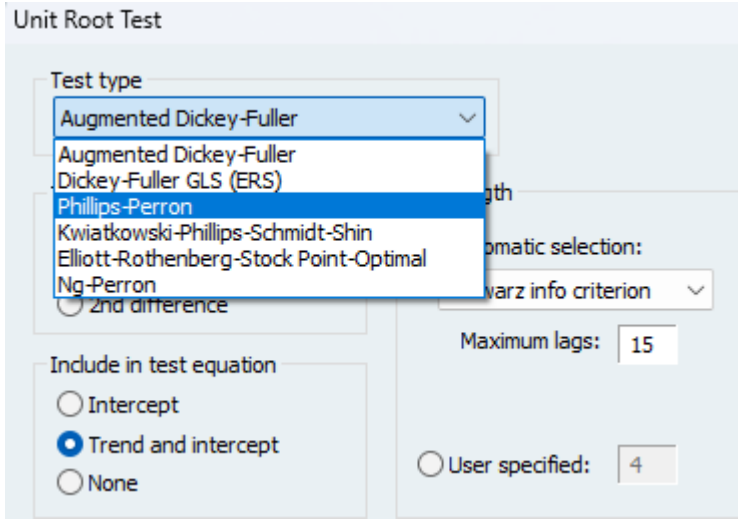
*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(CORE_CPI)
Method: Least Squares
Date: 05/07/26 Time: 21:42
Sample: 2005M01 2026M04
Included observations: 256

Variable	Coefficient	Std. Error	t-Statistic	Prob.
CORE_CPI(-1)	-0.002738	0.002003	-1.367098	0.1728
D(CORE_CPI(-1))	0.502501	0.061597	8.157862	0.0000
D(CORE_CPI(-2))	0.148724	0.069369	2.143959	0.0330
D(CORE_CPI(-3))	0.219693	0.062693	3.504270	0.0005
C	0.132409	0.090737	1.459270	0.1457
@TREND("2005M01")	0.000659	0.000392	1.679782	0.0942

R-squared	0.664025	Mean dependent var	0.214009
Adjusted R-squared	0.657306	S.D. dependent var	0.271326
S.E. of regression	0.158835	Akaike info criterion	-0.818748
Sum squared resid	6.307107	Schwarz criterion	-0.735658
Log likelihood	110.7998	Hannan-Quinn criter.	-0.785330
F-statistic	98.82067	Durbin-Watson stat	2.026467
Prob(F-statistic)	0.000000		

Lithuania CPI. PP test



Null Hypothesis: CORE_CPI has a unit root
Exogenous: Constant
Bandwidth: 11 (Newey-West automatic) using Bartlett kernel

		Adj. t-Stat	Prob.*
Phillips-Perron test statistic		1.868070	0.9998
Test critical values:	1% level	-3.455786	
	5% level	-2.872630	
	10% level	-2.572754	

*MacKinnon (1996) one-sided p-values.

Residual variance (no correction)	0.064339
HAC corrected variance (Bartlett kernel)	0.476036

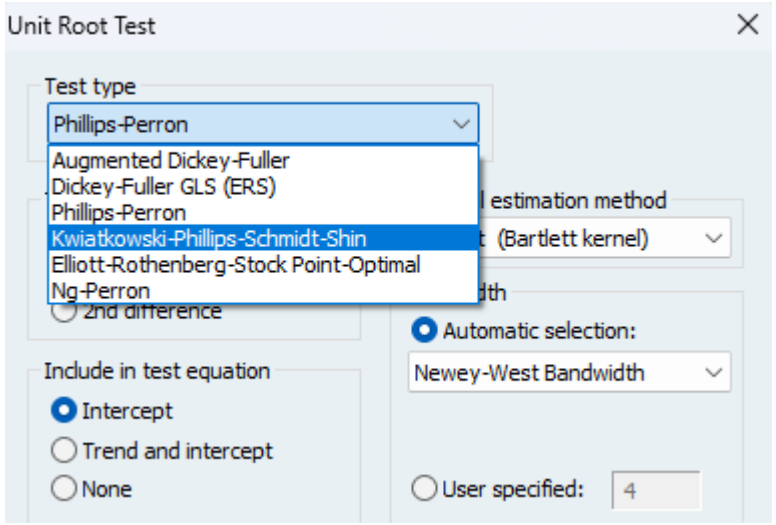
Phillips-Perron Test Equation
Dependent Variable: D(CORE_CPI)
Method: Least Squares
Date: 05/07/26 Time: 21:44
Sample: 2005M01 2026M04
Included observations: 256

Variable	Coefficient	Std. Error	t-Statistic	Prob.
CORE_CPI(-1)	0.006402	0.001075	5.958030	0.0000
C	-0.224043	0.075226	-2.978275	0.0032

R-squared	0.122620	Mean dependent var	0.214009
Adjusted R-squared	0.119165	S.D. dependent var	0.271326
S.E. of regression	0.254647	Akaike info criterion	0.109906
Sum squared resid	16.47068	Schwarz criterion	0.137603
Log likelihood	-12.06800	Hannan-Quinn criter.	0.121046
F-statistic	35.49812	Durbin-Watson stat	0.494173
Prob(F-statistic)	0.000000		

Fail to reject the null – series is nonstationary

Lithuania CPI. KPSS test



Null Hypothesis: CORE_CPI is stationary
Exogenous: Constant, Linear Trend
Bandwidth: 12 (Newey-West automatic) using Bartlett kernel

		LM-Stat.
Kwiatkowski-Phillips-Schmidt-Shin test statistic		0.379932
Asymptotic critical values*:	1% level	0.216000
	5% level	0.146000
	10% level	0.119000

*Kwiatkowski-Phillips-Schmidt-Shin (1992, Table 1)

Residual variance (no correction)	26.35311
HAC corrected variance (Bartlett kernel)	321.5872

KPSS Test Equation
Dependent Variable: CORE_CPI
Method: Least Squares
Date: 05/07/26 Time: 21:46
Sample: 2005M01 2026M04
Included observations: 256

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	44.48697	0.642330	69.25879	0.0000
@TREND("2005M01")	0.189392	0.004359	43.45197	0.0000

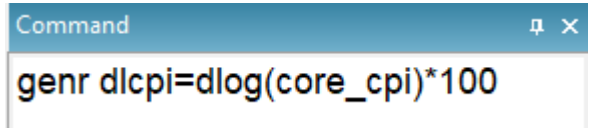
R-squared	0.881423	Mean dependent var	68.63450
Adjusted R-squared	0.880956	S.D. dependent var	14.93709
S.E. of regression	5.153699	Akaike info criterion	6.125088
Sum squared resid	6746.395	Schwarz criterion	6.152785
Log likelihood	-782.0113	Hannan-Quinn criter.	6.136228
F-statistic	1888.074	Durbin-Watson stat	0.002794
Prob(F-statistic)	0.000000		

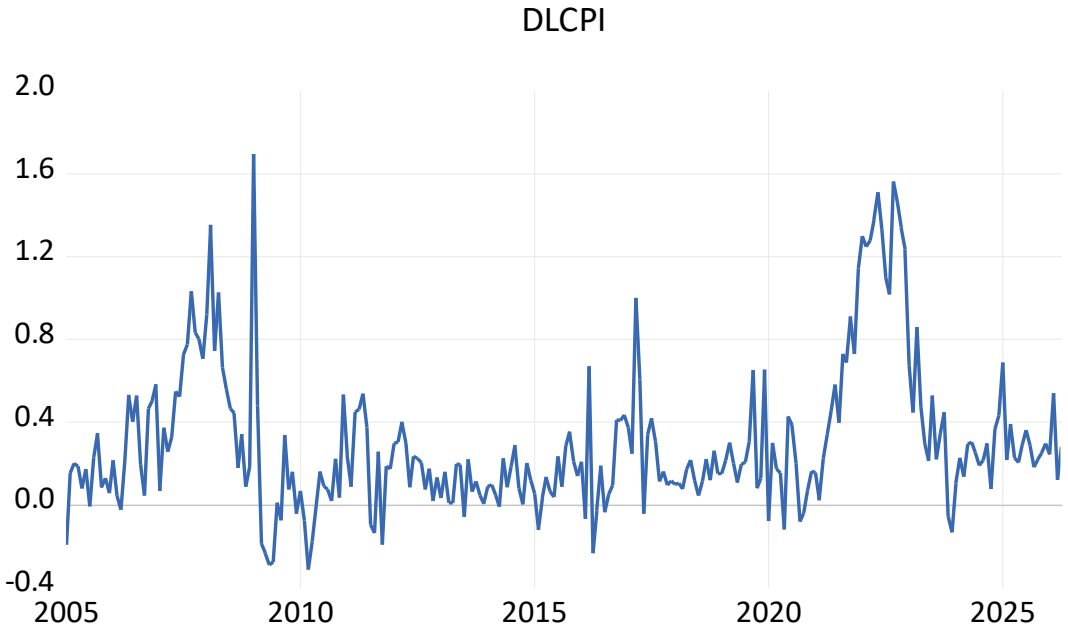
In KPSS test to reject the null the LM-statistic must be higher than the critical value.

We reject the null of trend stationarity at the 1% level → evidence of non-stationarity.

Lithuania CPI.
First difference (logs)

We take logs and then first differences of the CPI series by typing the following commands in the EViews command window.





The first difference looks stationary.

We multiply the first difference by 100 to get an approximation of the monthly growth rate in percent.

Lithuania CPI.
First difference (logs): Tests

Null Hypothesis: DLCPI has a unit root
Exogenous: Constant
Lag Length: 2 (Automatic - based on SIC, maxlag=15)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-3.586181	0.0067
Test critical values:	1% level	-3.455786	
	5% level	-2.872630	
	10% level	-2.572754	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(DLCPI)
Method: Least Squares
Date: 05/07/26 Time: 21:50
Sample: 2005M01 2026M04
Included observations: 256

Variable	Coefficient	Std. Error	t-Statistic	Prob.
DLCPI(-1)	-0.165279	0.046088	-3.586181	0.0004
D(DLCPI(-1))	-0.347398	0.065785	-5.280804	0.0000
D(DLCPI(-2))	-0.205988	0.061737	-3.336538	0.0010
C	0.050260	0.019992	2.514037	0.0126

R-squared	0.228512	Mean dependent var	0.000330
Adjusted R-squared	0.219327	S.D. dependent var	0.264313
S.E. of regression	0.233535	Akaike info criterion	-0.055467
Sum squared resid	13.74374	Schwarz criterion	-7.33E-05
Log likelihood	11.09974	Hannan-Quinn criter.	-0.033188
F-statistic	24.88047	Durbin-Watson stat	2.012289
Prob(F-statistic)	0.000000		

Null Hypothesis: DLCPI has a unit root
Exogenous: Constant
Bandwidth: 7 (Newey-West automatic) using Bartlett kernel

		Adj. t-Stat	Prob.*
Phillips-Perron test statistic		-6.590085	0.0000
Test critical values:	1% level	-3.455786	
	5% level	-2.872630	
	10% level	-2.572754	

*MacKinnon (1996) one-sided p-values.

Residual variance (no correction)	0.059936
HAC corrected variance (Bartlett kernel)	0.065004

Phillips-Perron Test Equation
Dependent Variable: D(DLCPI)
Method: Least Squares
Date: 05/07/26 Time: 21:51
Sample: 2005M01 2026M04
Included observations: 256

Variable	Coefficient	Std. Error	t-Statistic	Prob.
DLCPI(-1)	-0.277251	0.043350	-6.395708	0.0000
C	0.082864	0.020062	4.130330	0.0000

R-squared	0.138706	Mean dependent var	0.000330
Adjusted R-squared	0.135315	S.D. dependent var	0.264313
S.E. of regression	0.245780	Akaike info criterion	0.039023
Sum squared resid	15.34360	Schwarz criterion	0.066720
Log likelihood	-2.994934	Hannan-Quinn criter.	0.050162
F-statistic	40.90508	Durbin-Watson stat	2.348322
Prob(F-statistic)	0.000000		

Null Hypothesis: DLCPI is stationary
Exogenous: Constant
Bandwidth: 11 (Newey-West automatic) using Bartlett kernel

		LM-Stat.
Kwiatkowski-Phillips-Schmidt-Shin test statistic		0.238901
Asymptotic critical values*:	1% level	0.739000
	5% level	0.463000
	10% level	0.347000

*Kwiatkowski-Phillips-Schmidt-Shin (1992, Table 1)

Residual variance (no correction)	0.125529
HAC corrected variance (Bartlett kernel)	0.897365

KPSS Test Equation
Dependent Variable: DLCPI
Method: Least Squares
Date: 05/07/26 Time: 21:51
Sample: 2005M01 2026M04
Included observations: 256

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.298015	0.022187	13.43184	0.0000

R-squared	0.000000	Mean dependent var	0.298015
Adjusted R-squared	0.000000	S.D. dependent var	0.354995
S.E. of regression	0.354995	Akaike info criterion	0.770471
Sum squared resid	32.13543	Schwarz criterion	0.784320
Log likelihood	-97.62032	Hannan-Quinn criter.	0.776041
Durbin-Watson stat	0.549640		

Lithuania CPI. Suitable models

CPI is I(1). We use the first difference to build an ARIMA model.

ACF decays gradually. AR-type behavior

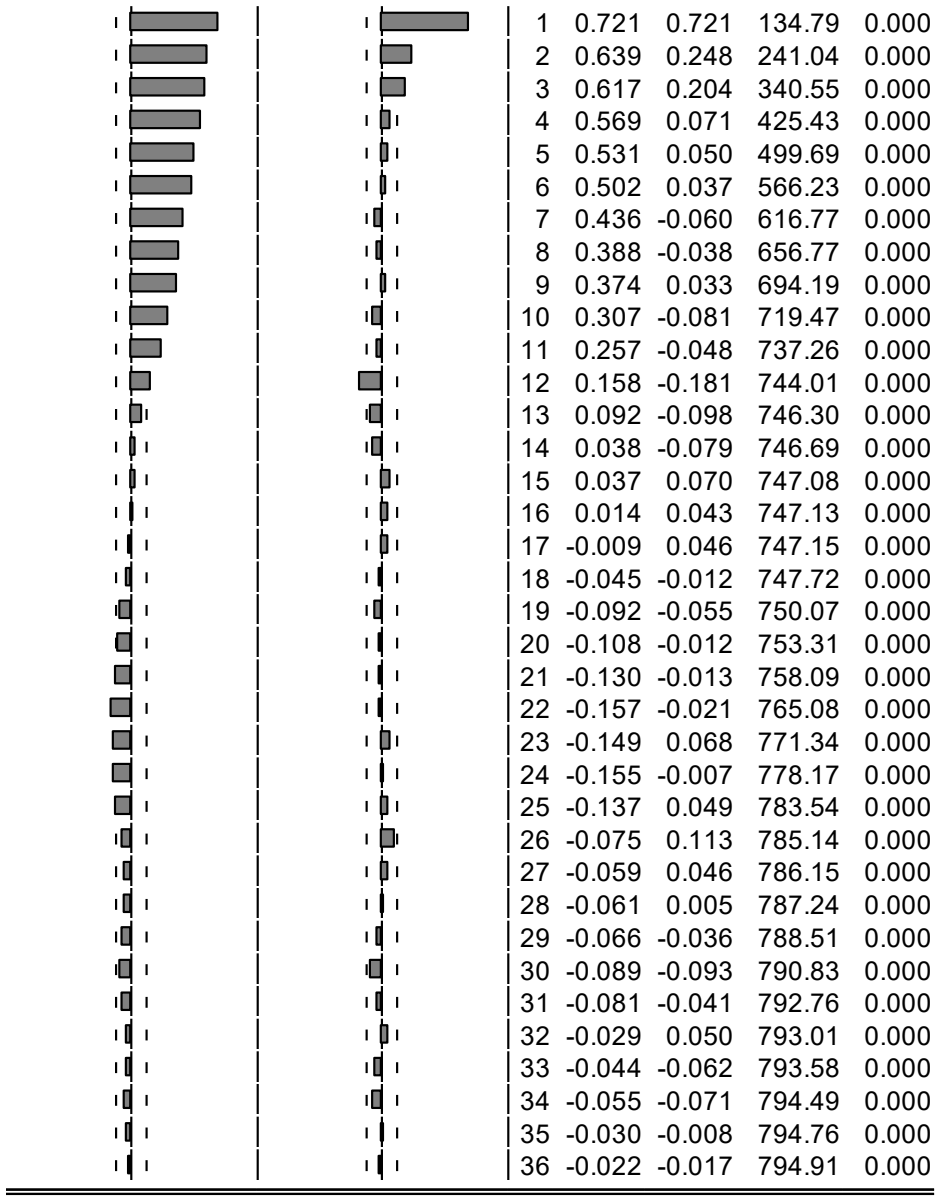
PACF dies down after lag 3.

The good first choice is ARIMA(3,1,0).

ARIMA(1,1,1) is also possible.

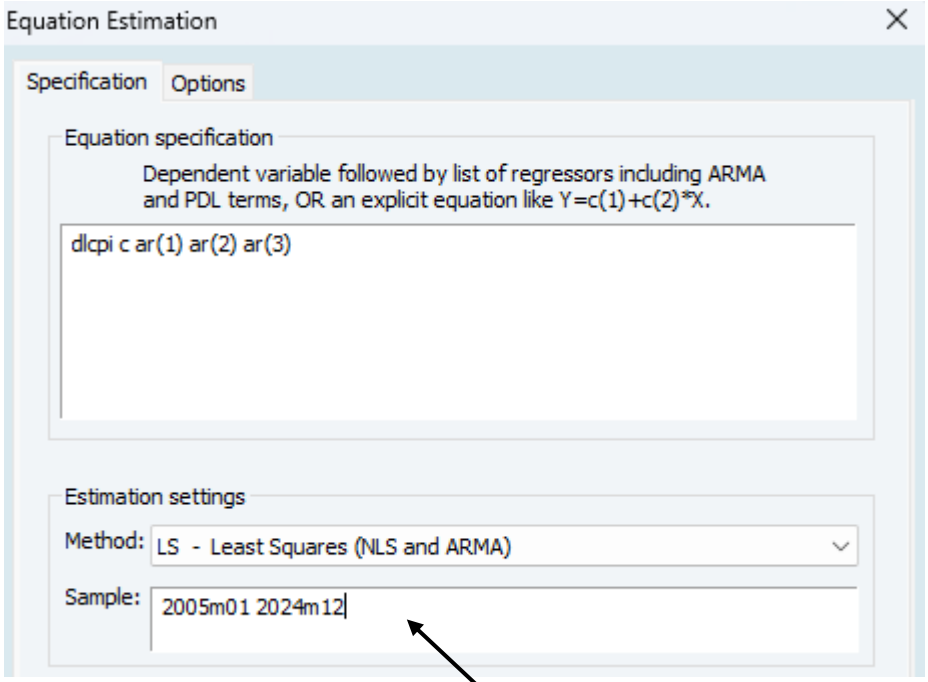
We should check it (especially if the residuals of ARIMA(3,1,0) have autocorrelation).

Date: 05/07/26 Time: 21:52
Sample: 2005M01 2026M04
Included observations: 256



Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.721	0.721	134.79	0.000
		2	0.639	0.248	241.04	0.000
		3	0.617	0.204	340.55	0.000
		4	0.569	0.071	425.43	0.000
		5	0.531	0.050	499.69	0.000
		6	0.502	0.037	566.23	0.000
		7	0.436	-0.060	616.77	0.000
		8	0.388	-0.038	656.77	0.000
		9	0.374	0.033	694.19	0.000
		10	0.307	-0.081	719.47	0.000
		11	0.257	-0.048	737.26	0.000
		12	0.158	-0.181	744.01	0.000
		13	0.092	-0.098	746.30	0.000
		14	0.038	-0.079	746.69	0.000
		15	0.037	0.070	747.08	0.000
		16	0.014	0.043	747.13	0.000
		17	-0.009	0.046	747.15	0.000
		18	-0.045	-0.012	747.72	0.000
		19	-0.092	-0.055	750.07	0.000
		20	-0.108	-0.012	753.31	0.000
		21	-0.130	-0.013	758.09	0.000
		22	-0.157	-0.021	765.08	0.000
		23	-0.149	0.068	771.34	0.000
		24	-0.155	-0.007	778.17	0.000
		25	-0.137	0.049	783.54	0.000
		26	-0.075	0.113	785.14	0.000
		27	-0.059	0.046	786.15	0.000
		28	-0.061	0.005	787.24	0.000
		29	-0.066	-0.036	788.51	0.000
		30	-0.089	-0.093	790.83	0.000
		31	-0.081	-0.041	792.76	0.000
		32	-0.029	0.050	793.01	0.000
		33	-0.044	-0.062	793.58	0.000
		34	-0.055	-0.071	794.49	0.000
		35	-0.030	-0.008	794.76	0.000
		36	-0.022	-0.017	794.91	0.000

Lithuania CPI. ARIMA(3,1,0)



We estimate the model on a shorter sample

Dependent Variable: DLCPI
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 08/03/26 Time: 14:17
Sample: 2005M01 2024M12
Included observations: 240
Convergence achieved after 26 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.288505	0.098993	2.914386	0.0039
AR(1)	0.505828	0.044577	11.34736	0.0000
AR(2)	0.117554	0.045542	2.581218	0.0105
AR(3)	0.213767	0.042684	5.008166	0.0000
SIGMASQ	0.055416	0.002781	19.92792	0.0000

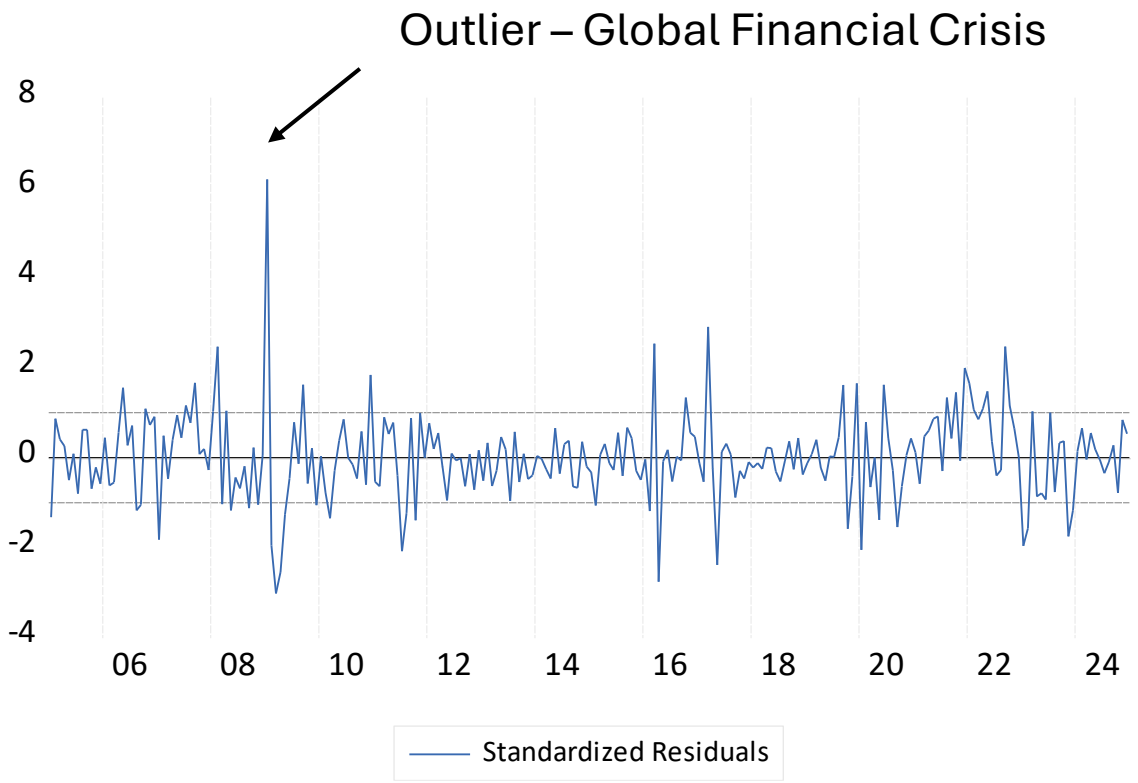
R-squared	0.582215	Mean dependent var	0.297953
Adjusted R-squared	0.575104	S.D. dependent var	0.364962
S.E. of regression	0.237897	Akaike info criterion	-0.009125
Sum squared resid	13.29983	Schwarz criterion	0.063388
Log likelihood	6.094982	Hannan-Quinn criter.	0.020093
F-statistic	81.87273	Durbin-Watson stat	2.021710
Prob(F-statistic)	0.000000		

Inverted AR Roots	.90	-.20+.45i	-.20-.45i

Coefficients are significant.

R-squared is sufficient.

Lithuania CPI. ARIMA(3,1,0). Residuals



Date: 08/03/26 Time: 14:19
Sample: 2005M01 2024M12
Q-statistic probabilities adjusted for 3 ARMA terms

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	-0.015	-0.015	0.0578	
		2	-0.025	-0.025	0.2126	
		3	-0.054	-0.055	0.9308	
		4	0.009	0.006	0.9489	0.330
		5	0.051	0.049	1.5888	0.452
		6	0.070	0.069	2.7924	0.425
		7	0.005	0.011	2.7985	0.592
		8	-0.016	-0.007	2.8609	0.721
		9	0.123	0.131	6.6842	0.351
		10	0.054	0.058	7.4329	0.385
		11	0.098	0.102	9.8799	0.274
		12	-0.060	-0.045	10.795	0.290
		13	-0.090	-0.086	12.848	0.232
		14	-0.128	-0.144	17.056	0.106
		15	-0.004	-0.053	17.061	0.147
		16	0.029	-0.011	17.286	0.187
		17	0.033	0.013	17.572	0.227
		18	-0.000	0.002	17.572	0.286
		19	-0.046	-0.029	18.127	0.317
		20	-0.032	-0.038	18.396	0.364
		21	-0.030	-0.028	18.638	0.414
		22	-0.110	-0.109	21.856	0.291
		23	-0.037	0.000	22.221	0.329
		24	-0.108	-0.087	25.367	0.232
		25	-0.085	-0.086	27.298	0.200
		26	0.078	0.052	28.957	0.182
		27	0.075	0.059	30.476	0.169
		28	0.029	0.047	30.700	0.199
		29	0.023	0.079	30.845	0.234
		30	-0.057	0.006	31.757	0.241
		31	-0.075	-0.026	33.304	0.225
		32	0.091	0.095	35.614	0.185
		33	-0.001	0.016	35.614	0.221
		34	-0.053	-0.060	36.398	0.232
		35	-0.000	-0.022	36.398	0.271
		36	0.004	-0.053	36.401	0.313

No visible autocorrelation in the residuals.

Lithuania CPI. ARIMA(1,1,1)

Dependent Variable: DLCPI
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 08/03/26 Time: 14:21
Sample: 2005M01 2024M12
Included observations: 240
Convergence achieved after 14 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.287021	0.106268	2.700909	0.0074
AR(1)	0.920391	0.028306	32.51533	0.0000
MA(1)	-0.466892	0.060248	-7.749505	0.0000
SIGMASQ	0.055738	0.002693	20.69491	0.0000

R-squared	0.579784	Mean dependent var	0.297953
Adjusted R-squared	0.574443	S.D. dependent var	0.364962
S.E. of regression	0.238082	Akaike info criterion	-0.011706
Sum squared resid	13.37722	Schwarz criterion	0.046305
Log likelihood	5.404676	Hannan-Quinn criter.	0.011668
F-statistic	108.5388	Durbin-Watson stat	1.936298
Prob(F-statistic)	0.000000		

Inverted AR Roots	.92
Inverted MA Roots	.47

Coefficients are significant.

SBC is a bit lower than for ARIMA(3,1,0)

No autocorrelation in the residuals.

Date: 08/03/26 Time: 14:22
Sample: 2005M01 2024M12
Q-statistic probabilities adjusted for 2 ARMA terms

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.027	0.027	0.1764	
		2	-0.089	-0.090	2.1231	
		3	0.037	0.042	2.4589	0.117
		4	0.004	-0.006	2.4631	0.292
		5	0.031	0.039	2.7070	0.439
		6	0.077	0.074	4.1749	0.383
		7	-0.003	-0.001	4.1766	0.524
		8	-0.012	-0.001	4.2135	0.648
		9	0.117	0.112	7.6292	0.366
		10	0.055	0.047	8.3826	0.397
		11	0.077	0.093	9.8813	0.360
		12	-0.050	-0.061	10.522	0.396
		13	-0.102	-0.090	13.187	0.281
		14	-0.126	-0.152	17.290	0.139
		15	-0.014	-0.047	17.340	0.184
		16	0.021	-0.012	17.451	0.233
		17	0.021	0.023	17.569	0.286
		18	-0.007	-0.001	17.582	0.349
		19	-0.057	-0.041	18.429	0.362
		20	-0.032	-0.036	18.700	0.411
		21	-0.047	-0.048	19.274	0.439
		22	-0.119	-0.110	23.071	0.285
		23	-0.039	0.007	23.470	0.319
		24	-0.111	-0.100	26.782	0.220
		25	-0.095	-0.070	29.242	0.172
		26	0.084	0.054	31.174	0.149
		27	0.070	0.052	32.493	0.144
		28	0.019	0.054	32.591	0.174
		29	0.040	0.081	33.042	0.196
		30	-0.059	-0.011	34.009	0.201
		31	-0.077	-0.028	35.654	0.184
		32	0.098	0.089	38.335	0.141
		33	0.002	0.003	38.337	0.171
		34	-0.063	-0.054	39.446	0.171
		35	0.006	-0.035	39.456	0.204
		36	0.019	-0.046	39.557	0.236

Lithuania CPI. ARIMA(1,1,1). Adding dummy

Dependent Variable: DLCPI
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 08/03/26 Time: 14:25
Sample: 2005M01 2024M12
Included observations: 240
Convergence achieved after 10 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.278922	0.108289	2.575726	0.0106
D2009M1	1.514465	0.243627	6.216315	0.0000
AR(1)	0.933351	0.022372	41.72023	0.0000
MA(1)	-0.450238	0.059703	-7.541260	0.0000
SIGMASQ	0.043465	0.003414	12.73079	0.0000

R-squared	0.672317	Mean dependent var	0.297953
Adjusted R-squared	0.666740	S.D. dependent var	0.364962
S.E. of regression	0.210688	Akaike info criterion	-0.251325
Sum squared resid	10.43151	Schwarz criterion	-0.178812
Log likelihood	35.15899	Hannan-Quinn criter.	-0.222107
F-statistic	120.5392	Durbin-Watson stat	2.013565
Prob(F-statistic)	0.000000		

Inverted AR Roots	.93
Inverted MA Roots	.45

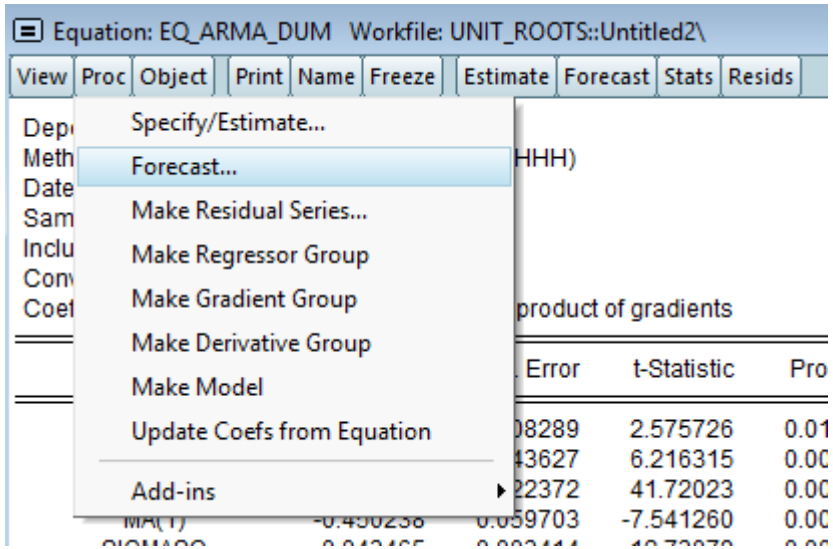
No significant changes

Date: 08/03/26 Time: 14:26
Sample: 2005M01 2024M12
Q-statistic probabilities adjusted for 2 ARMA terms

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob*
		1	-0.012	-0.012	0.0360	
		2	-0.026	-0.027	0.2073	
		3	0.055	0.054	0.9487	0.330
		4	0.066	0.067	2.0117	0.366
		5	0.055	0.060	2.7656	0.429
		6	0.078	0.081	4.2725	0.370
		7	0.019	0.019	4.3658	0.498
		8	-0.079	-0.086	5.9253	0.432
		9	0.097	0.080	8.2819	0.308
		10	0.082	0.066	9.9648	0.268
		11	0.012	0.018	10.004	0.350
		12	-0.089	-0.094	12.001	0.285
		13	-0.094	-0.115	14.271	0.218
		14	-0.105	-0.132	17.116	0.145
		15	0.010	-0.015	17.141	0.193
		16	-0.024	-0.031	17.295	0.241
		17	0.000	0.048	17.295	0.302
		18	-0.021	0.025	17.407	0.360
		19	-0.052	-0.028	18.122	0.381
		20	-0.027	-0.032	18.311	0.435
		21	-0.051	-0.054	18.989	0.458
		22	-0.080	-0.067	20.685	0.416
		23	-0.123	-0.090	24.760	0.258
		24	-0.062	-0.057	25.798	0.260
		25	-0.121	-0.127	29.735	0.157
		26	0.060	0.048	30.722	0.162
		27	0.043	0.050	31.216	0.182
		28	0.043	0.097	31.734	0.202
		29	0.052	0.121	32.491	0.214
		30	-0.062	-0.028	33.570	0.215
		31	-0.054	-0.060	34.387	0.225
		32	0.031	-0.003	34.661	0.255
		33	0.078	0.043	36.387	0.232
		34	-0.054	-0.047	37.222	0.241
		35	0.032	-0.002	37.518	0.270
		36	-0.012	-0.080	37.561	0.309

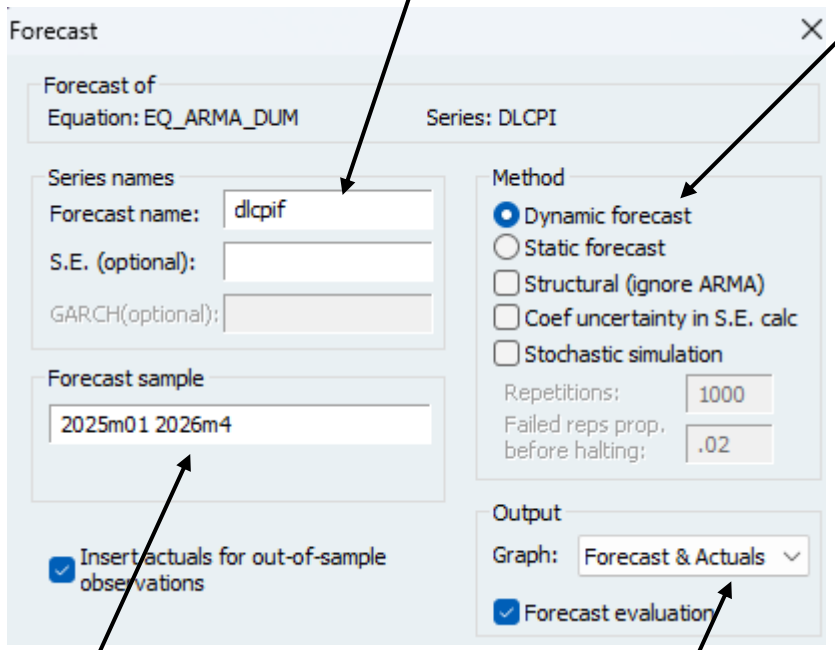
*Probabilities may not be valid for this equation specification.

Lithuania CPI. Forecast (1)



Series for the mean forecast

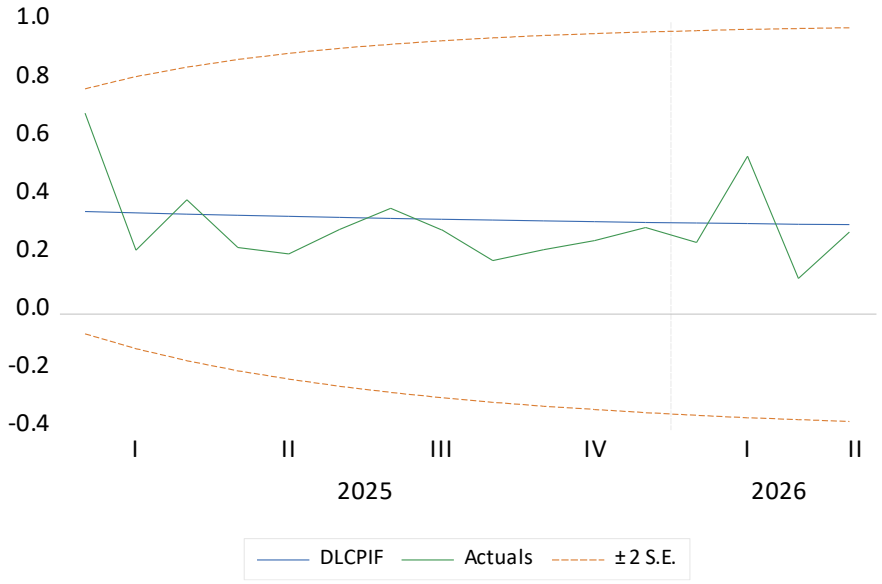
Use dynamic for iterative forecast



Specify the forecast range

Comparison with actual dynamics

Lithuania CPI. Forecast (2)



Forecast evaluation

Forecast: DLCPIF	
Actual: DLCPI	
Forecast sample: 2025M01 2026M04	
Included observations: 16	
Root Mean Squared Error	0.135423
Mean Absolute Error	0.106104
Mean Abs. Percent Error	41.27435
Theil Inequality Coef.	0.207497
Bias Proportion	0.032708
Variance Proportion	0.827711
Covariance Proportion	0.139581
Theil U2 Coefficient	0.634323
Symmetric MAPE	33.97863

$$RMSE = \sqrt{\frac{1}{n}\sum_{t=1}^{n}(\hat{y}_t - y_t)^2}$$

where $\hat{y}_t$ is the forecast, y_t is the actual, and n is the number of forecast periods.

RMSE (Root Mean Squared Error) is the typical size of the forecast error, in the **units of the variable itself.**

RMSE is most useful for **comparing rival models** on the same variable and sample

Theil's inequality coefficient (0.21) shows a reasonably accurate forecast, and its decomposition is the key: **bias is negligible (0.03)** – no systematic over/under-prediction – but the error is **dominated by the variance proportion (0.83)**, meaning the forecast is too smooth and doesn't track inflation's swings. **Theil's U2 (0.63 < 1)** confirms the model still forecasts better than a naïve no-change rule.

Summary.
ARIMA models estimation procedure

1. **Look at the series in levels.** Note trend, breaks, seasonality, and any change in volatility. Decide here whether logs are appropriate.
2. **Read the ACF and PACF of the levels.** Slow, near-linear decay in the ACF is the signature of a unit root.
3. **Test.** ADF and PP against a unit-root null, KPSS against a stationarity null, with deterministic terms chosen at step 1. If a break is visible, test allowing for it or split the sample.
4. **If non-stationary, difference.** Repeat steps 2 and 3 on $\Delta\ln(y)$. Stop at the smallest d that delivers stationarity.
5. **Identify and estimate.** Read p and q from the correlograms of the stationary series, then estimate a small set of ARIMA(p,d,q) candidates on a common sample.
6. **Diagnose.** Longest lag significant; residual correlogram and Ljung–Box clean; inverted roots inside the unit circle; hold-out forecasts acceptable. Any failure sends you back to step 5.